

- Deformation of the soft board

3 (a) Gravitational field strength at a point is the gravitational force per unit mass [B1] on a small test mass placed at that point.



Tampines Meridian Junior College

2025 JC2 Preliminary Examination H2 Physics

- (b) Gravitational force provides for centripetal force of satellite. [B1]

By Newton's 2nd Law [B1], gravitational field strength must thus be equal to the centripetal acceleration (OR $mg = ma \rightarrow g = a$)

- (c)(i) If obey inverse square law, $g = \frac{k}{r^2} \rightarrow \lg(g) = -2\lg(r) + \lg k$ [B1]

Consider points (7.95, 1.20) and (8.55, 0)

$$\text{Gradient} = \frac{1.20 - 0}{7.95 - 8.55} = -2 \quad [\text{B1}] \text{ working (showing coordinates) and answer}$$

Therefore must obey the inverse square law.

- (c)(ii) For $r = 4.18 \times 10^8 \text{ m} \rightarrow \lg(r) = 8.62$

(note: should not round off to 2 s.f. and use 8.6 for finding corresponding $\lg g$ value as it will greatly affect the accuracy of the final answer).

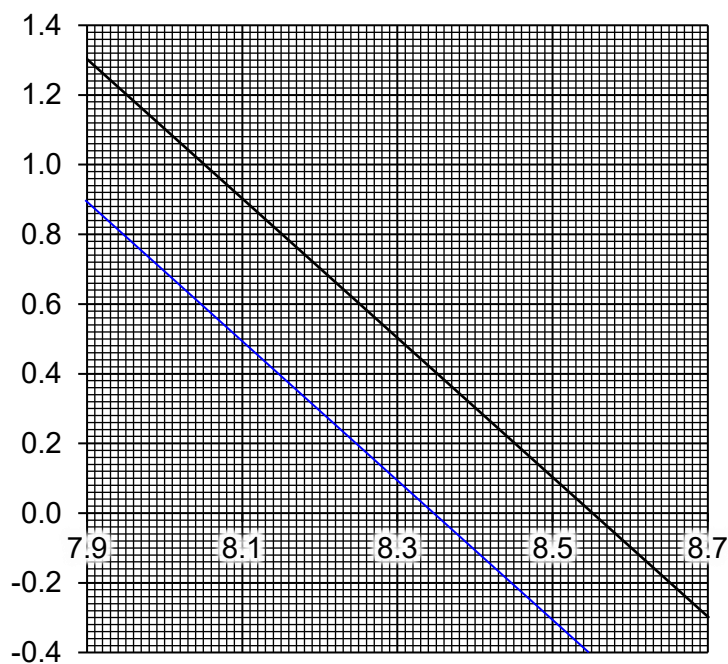
From graph, $\lg(g) = -0.14$ [B1]
 $g = 0.7244$

$$g = \frac{v^2}{r}$$

$$0.7244 = \frac{v^2}{4.18 \times 10^8} \quad [\text{C1}]$$

$$v = 17400 \text{ m s}^{-1} \quad [\text{A1}]$$

- (c)(iii)



Same gradient, lower than original graph (of planet X) [B1]

